UNIVERSITATEA DIN BUCUREȘTI
FACULTATEA DE MATEMATICĂ ȘI INFORMATICĂ

TUTORIAT
CALCUL NUMERIC
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Factorizarea QR prin metoda Householder

Exemplu:

$$Sa\ se\ scrie\ matricea\ A = \begin{pmatrix} 12 & -51 & 4 \\ 6 & 167 & -68 \\ -4 & 24 & -41 \end{pmatrix} sub\ forma\ A = Q * R, unde$$

$$Q\ este\ o\ matrice\ ortogonala\ respectiv\ R\ una\ superior\ triunghiulara\ prin\ metoda\ Householder.$$

Rezolvare:

$Vrem\ sa\ scriem\ R = H_1 H_0 A\ si\ Q = H_0 H_1\ (vezi\ curs)$

$$Alegem\ a_0 = \begin{pmatrix} 12 \\ 6 \\ -4 \end{pmatrix} prima\ coloana\ din\ A$$

$$\|a_0\|_2 = \sqrt{12^2 + 6^2 + (-4)^2} = \sqrt{196} = 14$$

$$\alpha_0 = sgn(a_{00}) * \|a_0\|_2 = sgn(12) * 14 = 1 * 14 = 14$$

$$v_0 = a_0 - a_0 e_0 = \begin{pmatrix} 12 \\ 6 \\ -4 \end{pmatrix} - \begin{pmatrix} 14 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ -4 \end{pmatrix}$$

$$H_0 = I_3 - 2\frac{v_0 {v_0}^t}{{v_0}^t v_0} \quad (1)$$

$Calculam\ {v_0}^t v_0$:

$$\begin{pmatrix} -2 & 6 & -4 \end{pmatrix} \begin{pmatrix} -2 \\ 6 \\ -4 \end{pmatrix} = (-2)(-2) + 6 * 6 + (-4)(-4) = 4 + 36 + 16 = 56$$

$$Relatia\ (1)\ devine: H_0 = I_3 - \frac{2}{56} v_0 {v_0}^t = I_3 - \frac{1}{28} v_0 {v_0}^t$$

$Calculam\ v_0 {v_0}^t$:

$$\begin{pmatrix}-2\\6\\-4\end{pmatrix}\begin{pmatrix}-2 & 6 & -4\end{pmatrix}=\begin{pmatrix}(-2)(-2) & -2*6 & (-2)(-4)\\6*(-2) & 6*6 & 6*(-4)\\(-4)(-2) & (-4)*6 & (-4)(-4)\end{pmatrix}=\begin{pmatrix}4 & -12 & 8\\-12 & 36 & -24\\8 & -24 & 16\end{pmatrix}$$

$$H_0=I_3-\begin{pmatrix}\frac{4}{28} & -\frac{12}{28} & \frac{8}{28}\\-\frac{12}{28} & \frac{36}{28} & -\frac{24}{28}\\\frac{8}{28} & -\frac{24}{28} & \frac{16}{28}\end{pmatrix}=I_3-\begin{pmatrix}\frac{1}{7} & -\frac{3}{7} & \frac{2}{7}\\-\frac{3}{7} & \frac{9}{7} & -\frac{6}{7}\\\frac{2}{7} & -\frac{6}{7} & \frac{4}{7}\end{pmatrix}=\begin{pmatrix}1-\frac{1}{7} & \frac{3}{7} & -\frac{2}{7}\\\frac{3}{7} & 1-\frac{9}{7} & \frac{6}{7}\\-\frac{2}{7} & \frac{6}{7} & 1-\frac{4}{7}\end{pmatrix}=$$

$$=\begin{pmatrix}\frac{6}{7} & \frac{3}{7} & -\frac{2}{7}\\\frac{3}{7} & -\frac{2}{7} & \frac{6}{7}\\-\frac{2}{7} & \frac{6}{7} & \frac{3}{7}\end{pmatrix}=\frac{1}{7}\begin{pmatrix}6 & 3 & -2\\3 & -2 & 6\\-2 & 6 & 3\end{pmatrix}=H_0$$

$Calculam\ H_0A$:

$$H_0A=\frac{1}{7}\begin{pmatrix}6 & 3 & -2\\3 & -2 & 6\\-2 & 6 & 3\end{pmatrix}\begin{pmatrix}12 & -51 & 4\\6 & 167 & -68\\-4 & 24 & -41\end{pmatrix}=$$

$$=\frac{1}{7}\begin{pmatrix}6*12+3*6+(-2)(-4) & 6*(-51)+3*167+(-2)*24 & 6*4+3*(-68)+(-2)(-41)\\3*12+(-2)*6+6*(-4) & 3*(-51)+(-2)*167+6*24 & 3*4+(-2)(-68)+6*(-41)\\-2*12+6*6+3*(-4) & (-2)(-51)++6*167+3*24 & -2*4+6*(-68)+3*(-41)\end{pmatrix}=$$

$$=\frac{1}{7}\begin{pmatrix}72+18+8 & -306+501-48 & 24-204+82\\36-12-24 & -153-334+144 & 12+136-246\\-24+36-12 & 102+1002+72 & -8-408-123\end{pmatrix}=$$

$$=\frac{1}{7}\begin{pmatrix}98 & 147 & -98\\0 & -343 & -98\\0 & 1176 & -539\end{pmatrix}=\begin{pmatrix}14 & 21 & -14\\0 & -49 & -14\\0 & 168 & -77\end{pmatrix}$$

$$Scriem\ H_0A\ pe\ blocuri\ i.e.:\ \begin{pmatrix}a_0 & \widetilde{A_{01}}\\0_{2,1} & \widetilde{A_{11}}\end{pmatrix}.\ Atunci\ \widetilde{A_{11}}=\begin{pmatrix}-49 & -14\\168 & -77\end{pmatrix}$$

$Continuam\ similar$:

$\widetilde{a_1}=\begin{pmatrix}-49\\168\end{pmatrix}\ prima\ coloana\ din\ \widetilde{A_{11}}$

$\|\widetilde{a_1}\|_2=\sqrt{(-49)^2+168^2}=175$

$\alpha_1=sgn(-49)*\|\widetilde{a_1}\|_2=sgn(-49)*175=-175$

$$\widetilde{v_1} = \widetilde{a_1} - \alpha_1 e_0 = \begin{pmatrix} -49 \\ 168 \end{pmatrix} - \begin{pmatrix} -175 \\ 0 \end{pmatrix} = \begin{pmatrix} 126 \\ 168 \end{pmatrix}$$

$$\widetilde{H_1} = I_2 - 2\frac{\widetilde{v_1}\widetilde{v_1}^t}{\widetilde{v_1}^t\widetilde{v_1}} \quad (2)$$

$Calculam\ \widetilde{v_1}^t\widetilde{v_1}$:

$$\begin{pmatrix} 126 & 168 \end{pmatrix}\begin{pmatrix} 126 \\ 168 \end{pmatrix} = 126^2 + 168^2 = 44100$$

$$Relatia\ (2) devine\ \widetilde{H_1} = I_2 - \frac{2}{44100}\widetilde{v_1}\widetilde{v_1}^t = I_2 - \frac{1}{22050}\widetilde{v_1}\widetilde{v_1}^t$$

$Calculam\ \widetilde{v_1}\widetilde{v_1}^t$:

$$\begin{pmatrix} 126 \\ 168 \end{pmatrix}\begin{pmatrix} 126 & 168 \end{pmatrix} = \begin{pmatrix} 126 * 126 & 126 * 168 \\ 168 * 126 & 168 * 168 \end{pmatrix} = \begin{pmatrix} 15876 & 21168 \\ 21168 & 28224 \end{pmatrix}$$

$$\widetilde{H_1} = I_2 - \frac{1}{22050}\begin{pmatrix} 15876 & 21168 \\ 21168 & 28224 \end{pmatrix} ☺ = \cdots = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \frac{18}{25} & \frac{24}{25} \\ \frac{24}{25} & \frac{32}{25} \end{pmatrix} = \begin{pmatrix} \frac{7}{25} & -\frac{24}{25} \\ -\frac{24}{25} & -\frac{7}{25} \end{pmatrix}$$

$H_1\ se\ obtine\ astfel\ (vezi\ curs)$:

$$H_1 = \begin{pmatrix} 1 & 0_{1,2} \\ 0_{2,1} & \widetilde{H_1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{7}{25} & -\frac{24}{25} \\ 0 & -\frac{24}{25} & -\frac{7}{25} \end{pmatrix}$$

$Ramane\ de\ aflat\ Q\ respectiv\ R$:

$R = H_1 H_0 A$

$Q = H_0 H_1$

$A = QR$

Exerciții propuse:

$$Acelasi\ enunt\ ca\ la\ exercitiul\ anterior\ dar\ A = \begin{pmatrix} 8 & 0 & -7 & 0 \\ 8 & 8 & 11 & 2 \\ 8 & -6 & 1 & -2 \\ -8 & -2 & -3 & -12 \end{pmatrix}$$